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Försättsblad tentamen / Examination cover

2707183

Anonymitetskod / Anonymous code		
S - 0003 - ZYP		
Kurskod / Course code: DT081A	Provkod/ Test code: T101	Tentamensdatum / Examination date: 2024-10-29
Kursnamn / Course name: Datateknik AV, Signal- och bildbehandling		
Provnamn / Test name: Tentamen		

Skriv din anonymitetskod på varje inlämnat papper
Write your anonymous code on each sheet submitted

Casio fx-cg 50

Sätt ett kryss (x) för varje inlämnad uppgift
Use an x to indicate which questions has been submitted

Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes	Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes
1	X	4,5		16			
2	–	0		17			
3	X	3		18			
4	X	4,5		19			
5	X	0		20			
6	X	3,5		21			
7	X	2,5		22			
8	X	2,5		23			
9	X	5		24			
10				25			
11				26			
12				27			
13				28			
14				29			
15				30			
Poängsumma / Points		25,5	Betyg / Grade: D	Lärarsign./ Teachers sign			

Fylls i av tentamensvakt / To be filled in by the invigilator

Antal lösa blad/ No. of sheets submitted	Inlämnad tentamen / Submitted exam	Leg kontroll / Control identification	Sign. tentamensvakt / Sign. invigilator
9	0	ok	J.W.

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Försättsbladet skall alltid lämnas in även om ingen uppgift behandlats
Examination cover should always be submitted even if no questions are answered

Utskriven 2024-10-28 kl. 12:37:16

Example Solutions for DT081A – Signal and Image Processing Exam, 2024-10-29

1 (1 + 1 + 1 + 1 + 1 = 5 p)

a)
The process of converting a continuous-time signal into a discrete-time signal by taking periodic samples, $x[n] = x(nT_s)$ where T_s is the sampling interval.

b)
The process of mapping a continuous range of values to a finite set of discrete levels, often in analog-to-digital conversion.

c)
The output of a system when an impulse signal is applied at its input. It characterizes the system's behavior.

d)
The steady-state response of a system to sinusoidal inputs at varying frequencies.

e)
A technique used to reduce spectral leakage in signal analysis by applying a window function to a signal before performing a Fourier transform.

2 (3 + 2 = 5 p)

a)
$y[n] = s[n] * h[n]$

$y[0] = (3)(-0.5) = -1.5$
$y[1] = (7)(-0.5) + (3)(0) = -3.5$
$y[2] = (10)(-0.5) + (7)(0) + (3)(0.5) = -5 + 1.5 = -3.5$
$y[3] = (3)(-0.5) + (10)(0) + (7)(0.5) = -1.5 + 3.5 = 2$
...

$y[n] = \{-1.5, -3.5, -3.5, 2, 7.5, 1.5, -2.5\}$

b) This filter approximates a differentiation operation, emphasizing changes in the signal, which is characteristic of a high-pass filter.

3 (1 + 2 = 3 p)

a) The system diagram shows both a feed-forward and feedback branch. The latter implies that the system will produce an infinite impulse response IIR.

b)
$y(n) = x(n) + 2x(n-1) + x(n-3) - 3y(n-1) + 2y(n-3)$

4 (2 + 3 = 5 p)

a)
Extend spectrum plot to at least $-2\pi < \Omega < 2\pi$, where the continous spectrum is mirrored on the y-axis and replicated over the whole Ω axis, with a periodicity of 2π.

b)
$f_s \geq 2f_{max}$
The lowest sampling frequency depends on the signal bandwidth. The figure shows that $f_{max} = \frac{3}{4}\pi\frac{f_s}{2\pi} = \frac{3}{8}f_s = \frac{150}{8} = 18.75Hz$. Hence $f_s \geq 37.5Hz$.

5 (2 + 4 = 6 p)

a) Amplitude and phase relationship:
$|Y(f)| = |H(f)||X(f)|$
$\angle Y(f) = \angle H(f) + \angle X(f)$

b) Sketch spectra

$$|Y(f)| = \begin{cases} 8f & \text{for } 0 \leq f \leq 10 \\ 80 - 4(f-10) & \text{for } 10 < f \leq 30 \\ 0 & \text{for } f < 0 \text{ or } f > 30 \end{cases}$$

and

$arg(Y(f)) = 0 \forall f$

6 (2 + 2 = 4 p)

a) Window iii) has the narrowest main lobe, which provides the best frequency separation between closely spaced frequency components allowing for better distinction.
b) Window iv) has zero outside of main lobe, which effectively eliminates spectral leakage in the frequency domain ensuring that frequency components are not contaminated by artifacts from distant frequencies.

7 (3 p)

At low frequencies the system has an amplification of 1, whereas higher frequencies are amplified more. Combined this results in high frequency amplification (not to be confused with a high pass filter), which if applied to an image would enhance edges.

8 (1 + 1 + 1 + 1 + 1 = 5 p)

- $1 \rightarrow ii$
  Only the center element is used, and reduce the input with 0.5.
- $2 \rightarrow iv$
  A vertical filter emphasized vertical edges only.
- $3 \rightarrow v$
  Edge detecting kernel with elements summing to 0, only shows edges.
- $4 \rightarrow iii$
  Avering kernel smooths the image as a low-pass filter.
- $5 \rightarrow i$
  Edge detecting kernel summed with a unit kernel translates kernel to one with center element = 5, with elements summing to 1.

9 (6 p)

$x(i,j)$ has several non-zero values, that each contribute to the $y(i,j)$ through convolution with $h(i,j)$. From examining the isolated scaled impulse responses we can determine that $h(i,j)$ has a cross-shaped pattern. By subtracting one scaled impulse reponse from another we can identify the impulse response as.

$$h(i,j) = \begin{bmatrix} 0 & 1/8 & 0 \\ 1/8 & 1/2 & 1/8 \\ 0 & 1/8 & 0 \end{bmatrix}$$

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Lärarens anteckning / Teachers note:

1. a) Sampling: Sampling is the process of turning a continues signal into a discrete signal by taking sample points from the continues signal with a predetermined frequency. $S(t) \Rightarrow S(n)$ 1

b) Quantization is a process of altering the quantization level or intensity values range in signal that is discrete. How many values can be represented. 1

c) The impulse response is a systems response/ output to an impulse input. 1

d) The frequency response is the response of the frequency in a system after a specific input. Used for filtering. 0,5

e) Windowing is the process of defining a window with specific size to analyze a signal in a certain window or range to analyze it. Good for repeating deterministic signals. 1

4,5

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Lärarens anteckning / Teachers note:

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Lärarens anteckning / Teachers note:

3. systems diffrence equation:

$$Y(n) = X(n) + 2X(n-1) + X(n-3) + (-3Y(n-1) + 2Y(n-3))$$

2

The system is Causal, only looking into Past and Present values not future.

Impulse response $h(n) = \delta(n) + 2\delta(n-1) + \delta(n-3) + -3h(n-1) + 2h(n-3)$

where $\delta(n) = 1$ when $n=0$ else 0

$h(0) = 1$

$h(1) = 2 - 3h(0) = 2 - 3 = -1$

$h(2) = -3h(1) = 3$

The system introduces feedback so it is an IIR system.

1

linearity $aX_1 = aY_1$

/3

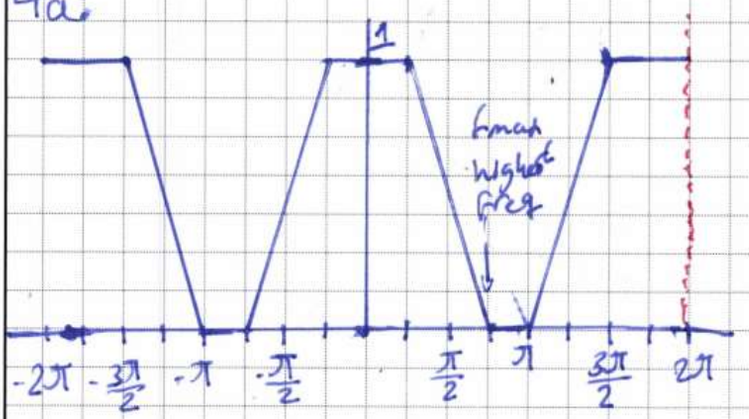
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Sidnr / Page no: 4

Lärarens anteckning / Teachers note:

4a.



1,5

b) lowest sampling freguency according to Nyquist shannon sampling theory should be $2 \cdot f_{max}$

$$\Omega = 2\pi \frac{F}{f_s} = \frac{2\pi F}{50000} \qquad f_{max} = \frac{3}{4}\pi$$

$$2\pi \frac{F}{50000} = \frac{3}{4}\pi \Rightarrow f_{max} = \frac{\frac{3}{4}\pi \cdot 50000}{2\pi} =$$

$$= 18750 \text{ Hz}$$

$f_{max} = 18750$ Hz

$f_s = 18750 \cdot 2 = 37500$ Hz

3

4,5

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5. the Phase is the phase-shift of the signal and the amplitude is the angle.

Lärarens anteckning / Teachers note:

/0

Lärarens anteckning / Teachers note:

6 a) iii) Picture 3 has the best frequency seperation as can be seen by the long height, narrow width of the main frequency meaning it takes in less of the surrounding frequencies in the main frequency component.

2

iv)

b) Picture 4 shows the best windowing function when it comes to spectral leakage where we can see in the x axis that we have 1 frequency component showing and no small spikes throughout the spectrum.

1,5

3,5

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Lärarens anteckning / Teachers note:

7. The system probably have some kind of High pass filtering We can see that the middle part corresponding to the lower frequency in a spectogram is virtually 0 meaning the low frequencies are missing while the higher frequencies (corners) have the highest values meaning it is a high frequency image left either through filter or just a highly detailed image.

2,5

Lärarens anteckning / Teachers note:

8. Output = $aI(x,y) + b$
a = brightness b = contrast.

i) = 4) When convolving the kernel with the imag we addt $\frac{1}{9}$ • value to the pixel increasing the contrast. as seen in the formula above. 0

ii) = 1) by multiplying each pixel by 0.5 we decrese the brightness. as seen in the formula above. 1

iii) = 2) the image is blurry with less edges telling us it has been filtered with low pass filter. kernel 2 takes away the current filtered pixel values to the left removing the high frequencies. 0,5

iv) = 3) Picture 4 is a high pass filtered with the kernel multiplying the value with 4 and lowering neighboring pixels creating almost black spots in low frequency areas since their value are similar which means subtraction 2 similar values almost is 0 which is black, or darker. 0,5

V) = 5 Same as picture 4 but even higher cutoff frequency we can say creating less low frequencies and more high frequencies. 0,5

2,5

Lärarens anteckning / Teachers note:

9. Putting the middle of the kernel on a Pixel looks like having a factor 0.5.

K(1,1)

0	0.125	0
0.125	0.5	0.125
0	0.125	0

Values on the top corners are 0

When tried on Pixel(1,1) and there are no other Pixels affecting this value. the Pixels in position (0,1) In the kernel is 0.125 when looking at Y(0,1) and (1,0) since these values only get affected by the 112 Pixel in I(1,1)

for the bottom left corner we can observe Y(5,2) and we have 0, affected only by Y(4,3)

the right middle in the kernel K(1,2) can be observed in $y(2,4) = \frac{28}{224} = 0.125$

K(2,1) observed in $y(5,3) = \frac{22}{176}$

for K(2,2) we need to use values in K(i,j) we already know. We can observe Y(5,4) = 6 affected by X(5,5) and X(4,3)

we know that X(5,5) when convolved with the kernel will output X(5,5) * 0.125 = 6 meaning X(4,3) doesn't influence the result. hence K(2,2) = 0

5